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(54) **DUAL RING STUFFING BOX**

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(58) **Field of Classification Search**

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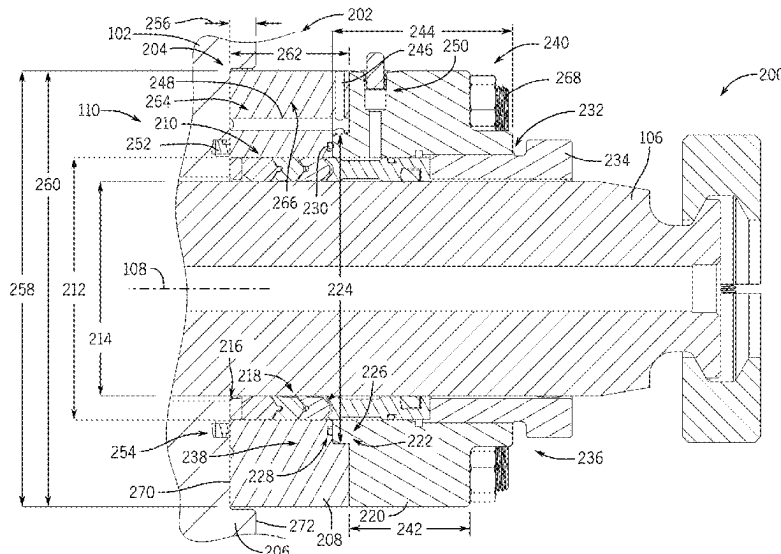
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(57) **ABSTRACT**

A system includes a fluid end having a fluid end face and a
recess extending toward a base. The system also includes a
pressure ring extending from the fluid end face, the pressure
ring having a pressure ring bore diameter larger than a fluid
end bore diameter. The system further includes an adjacent
ring coupled to the fluid end, the pressure ring positioned
between the adjacent ring and the base, wherein at least one
fastener extends through an aperture of the adjacent ring to
couple the adjacent ring to the fluid end. The system includes
packing material positioned within at least a portion of the
pressure ring bore, the packing material positioned such that
axial movement toward the fluid end is blocked by the base.

20 Claims, 14 Drawing Sheets



Related U.S. Application Data

continuation-in-part of application No. 17/213,854, filed on Mar. 26, 2021, now Pat. No. 11,391,374, which is a continuation-in-part of application No. 29/769,314, filed on Feb. 4, 2021, now Pat. No. Des. 933,104, and a continuation-in-part of application No. 29/769,318, filed on Feb. 4, 2021, now Pat. No. Des. 933,105.

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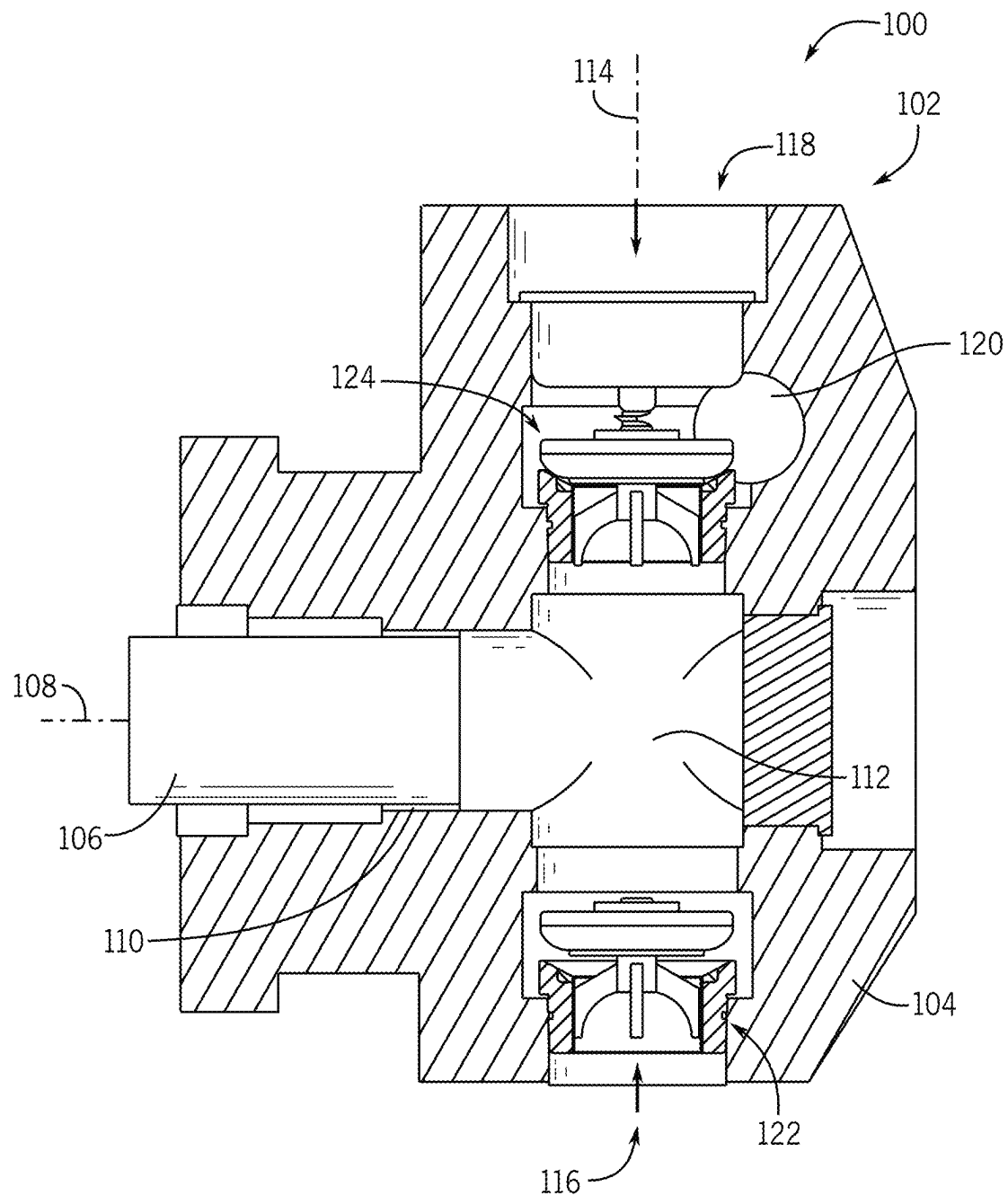


FIG. 1

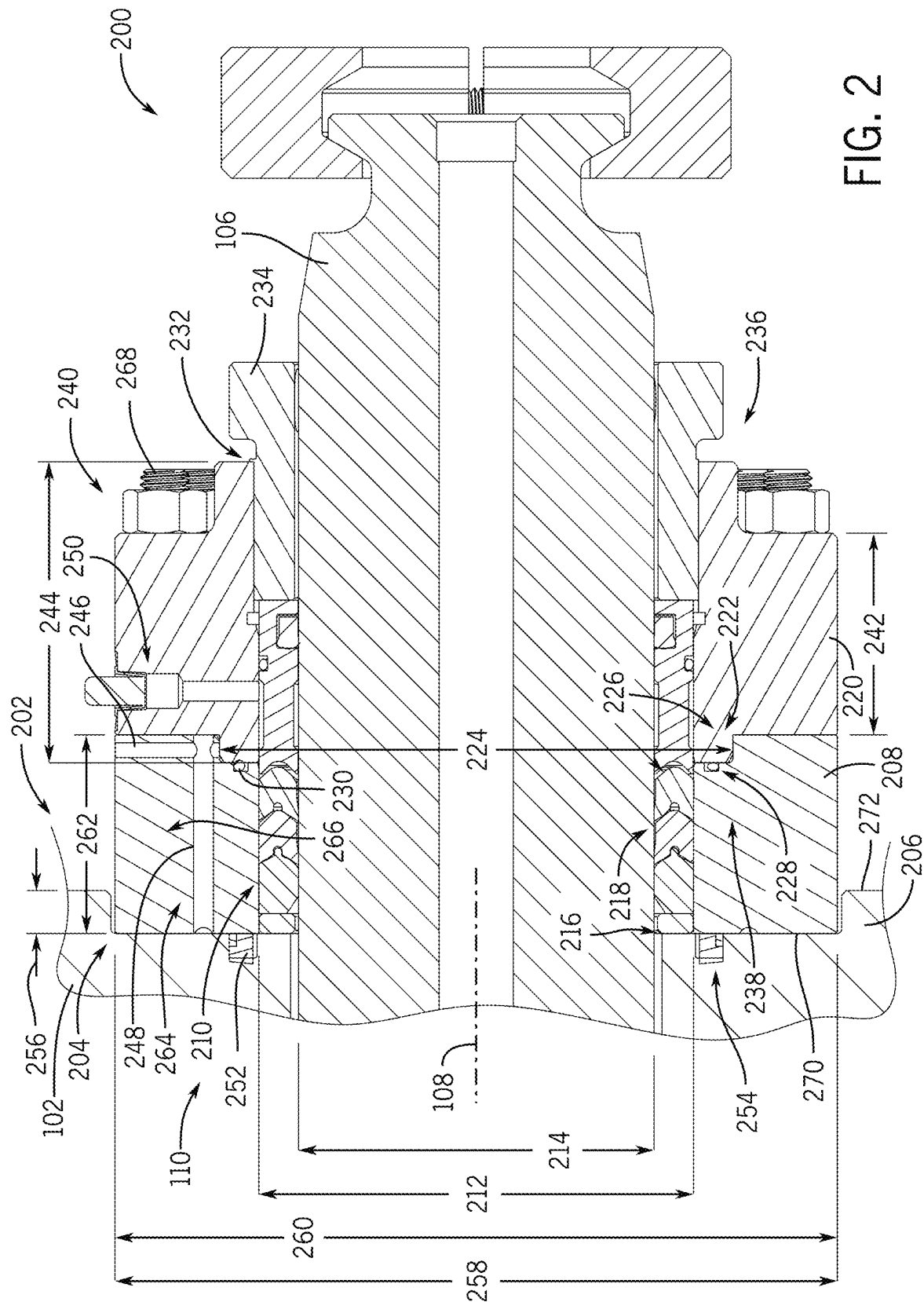


FIG. 2

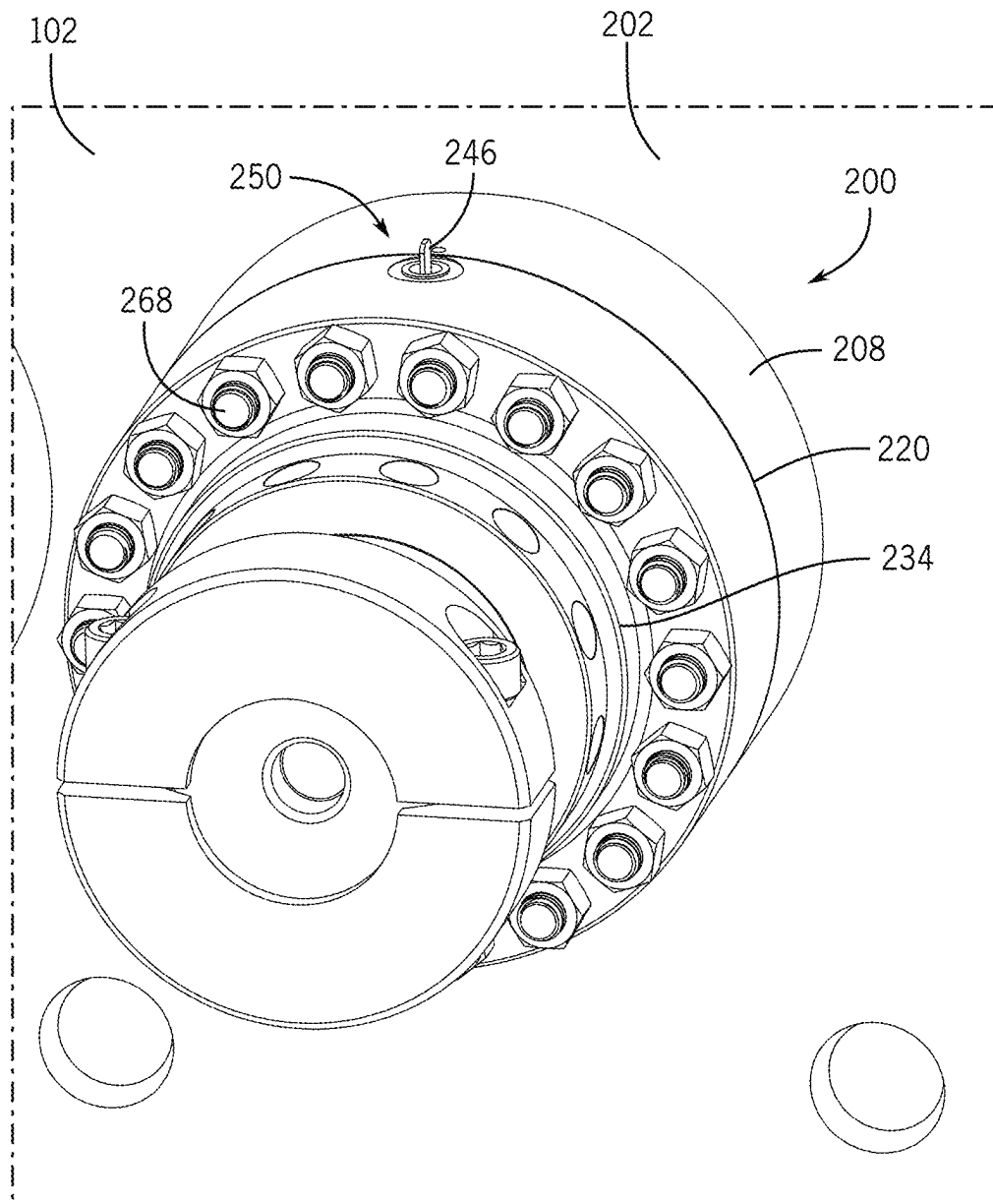
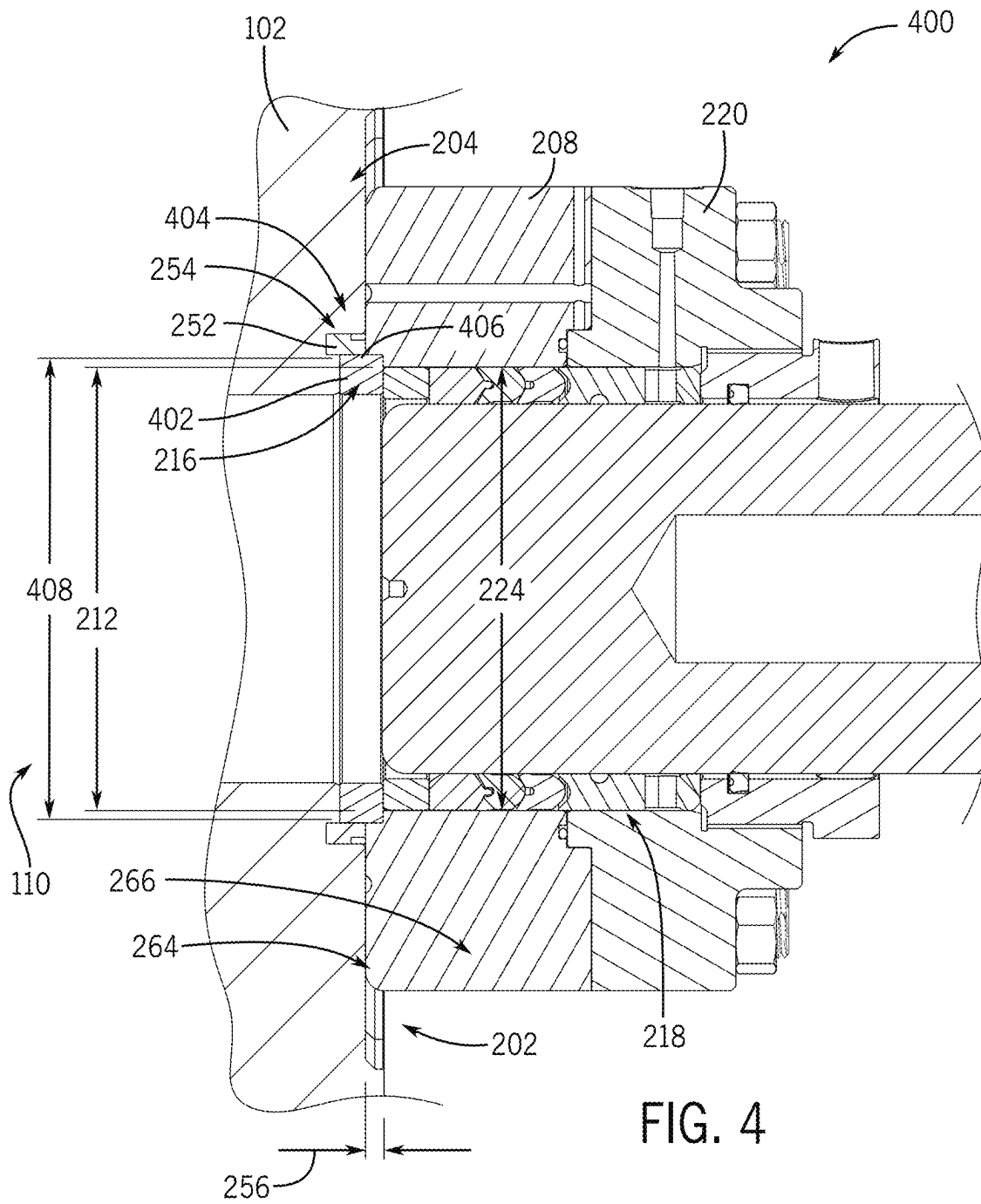


FIG. 3



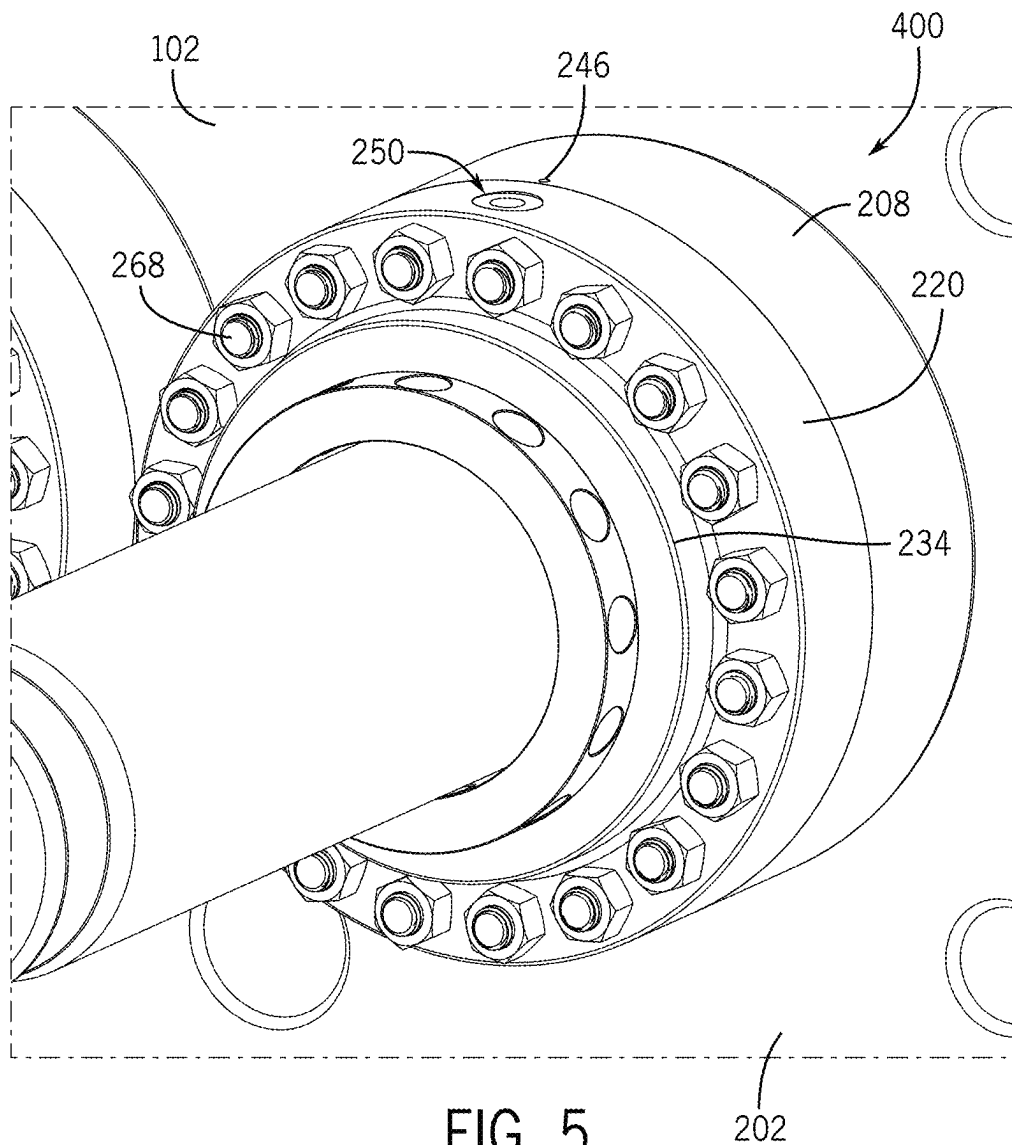


FIG. 5

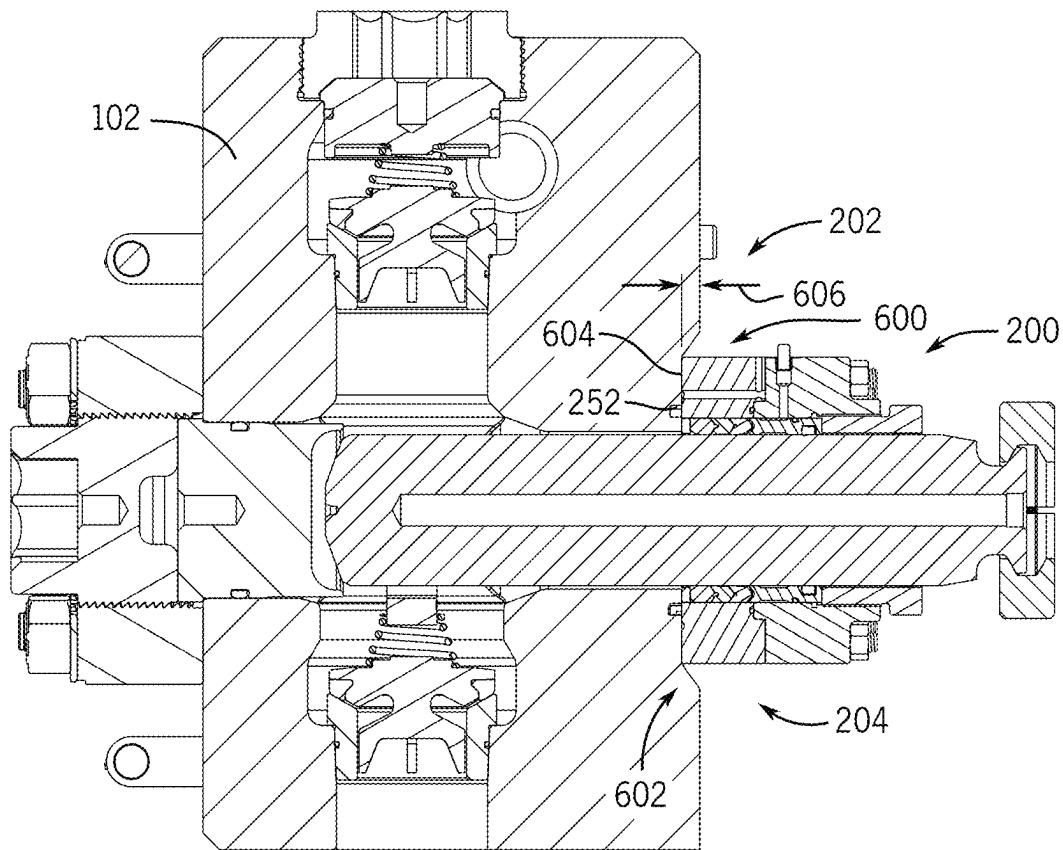
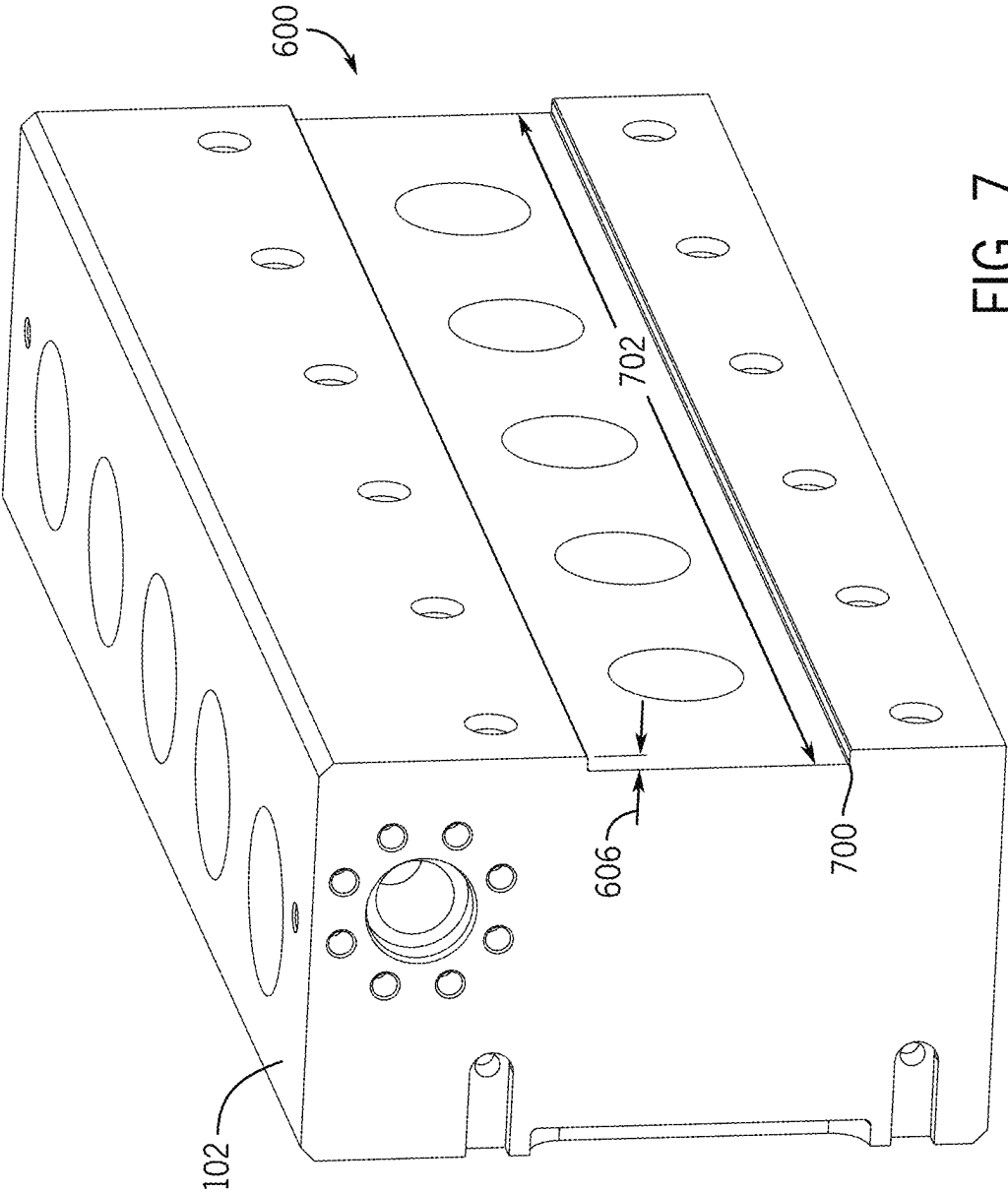


FIG. 6



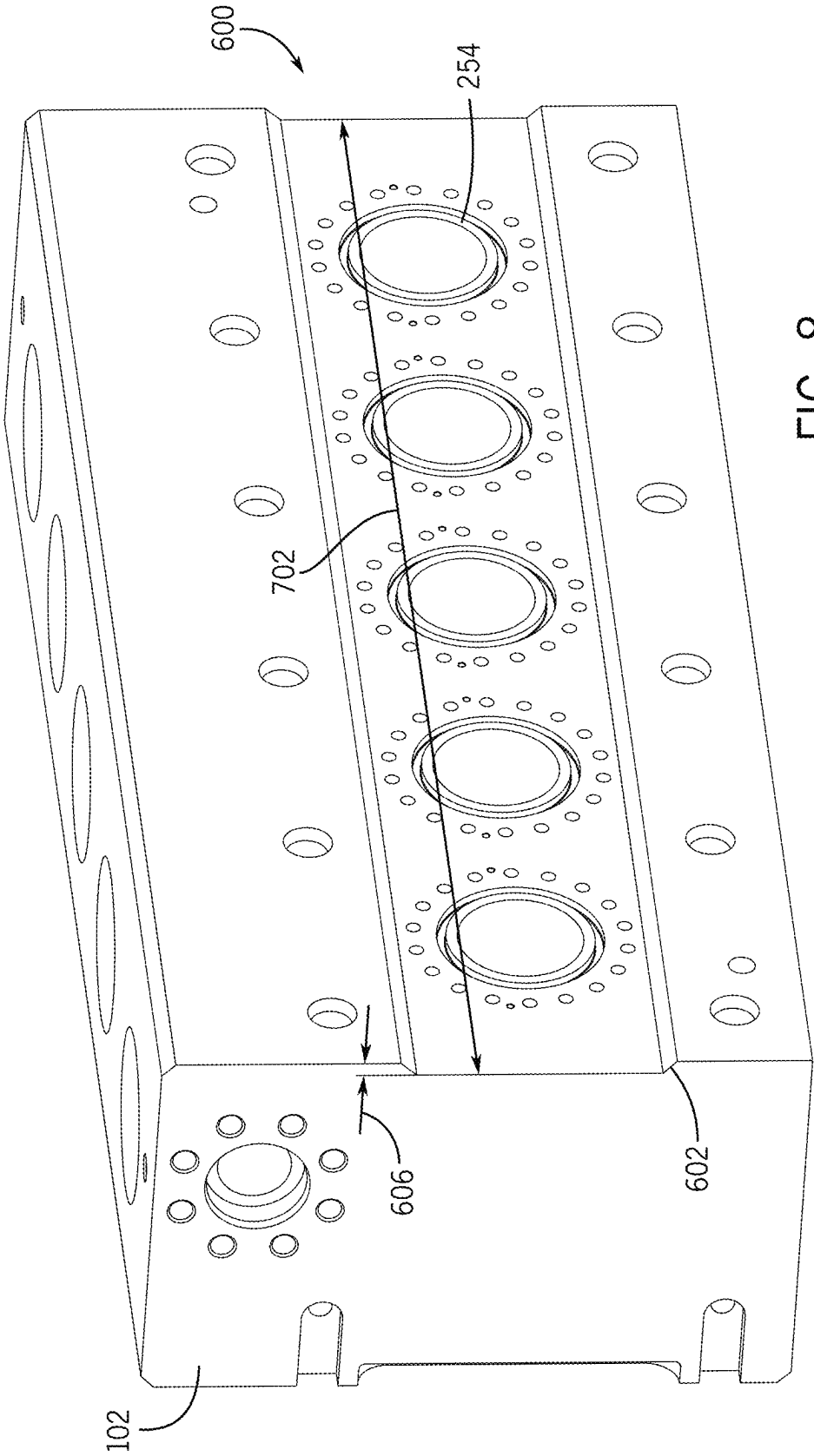
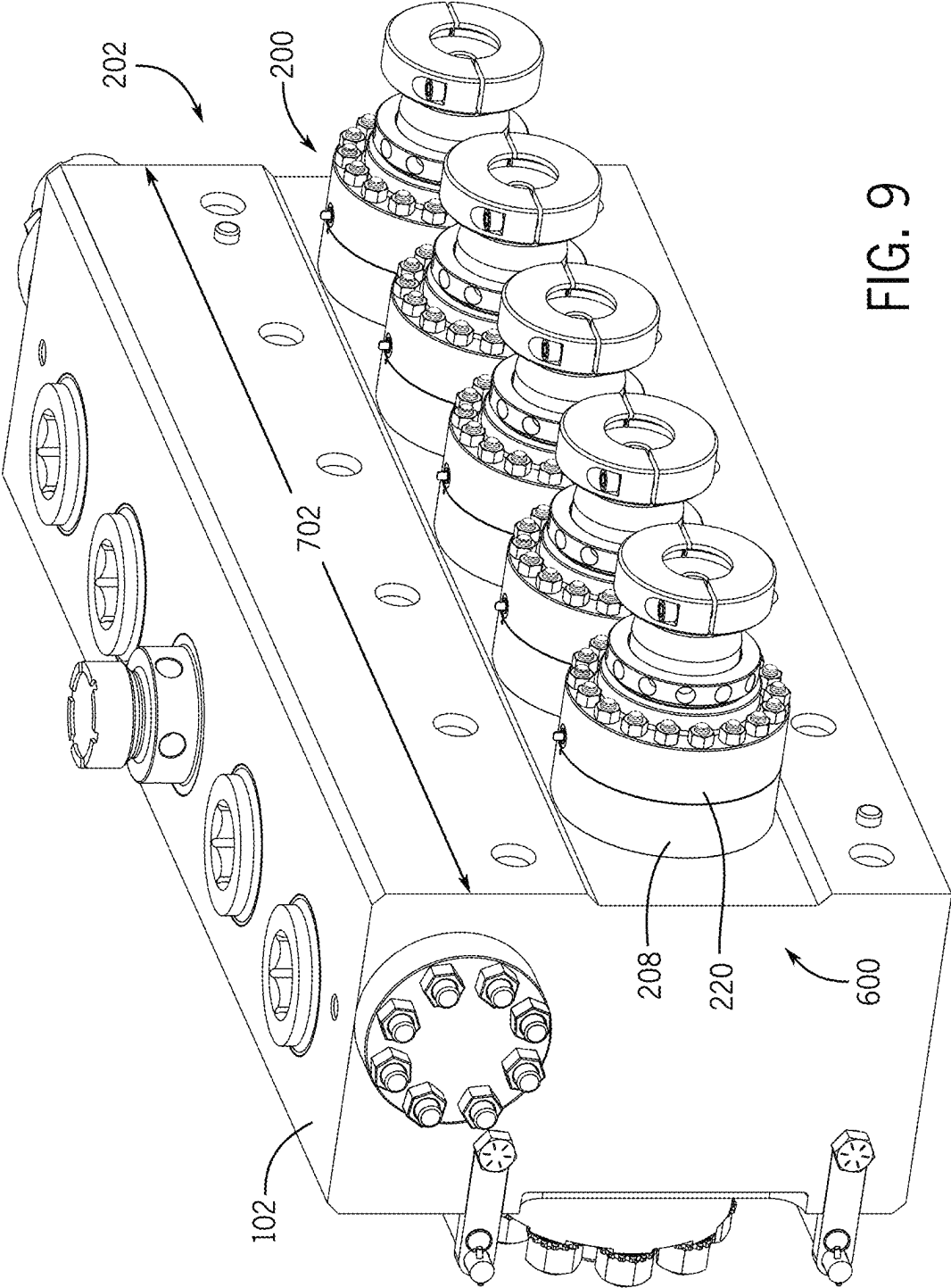


FIG. 8



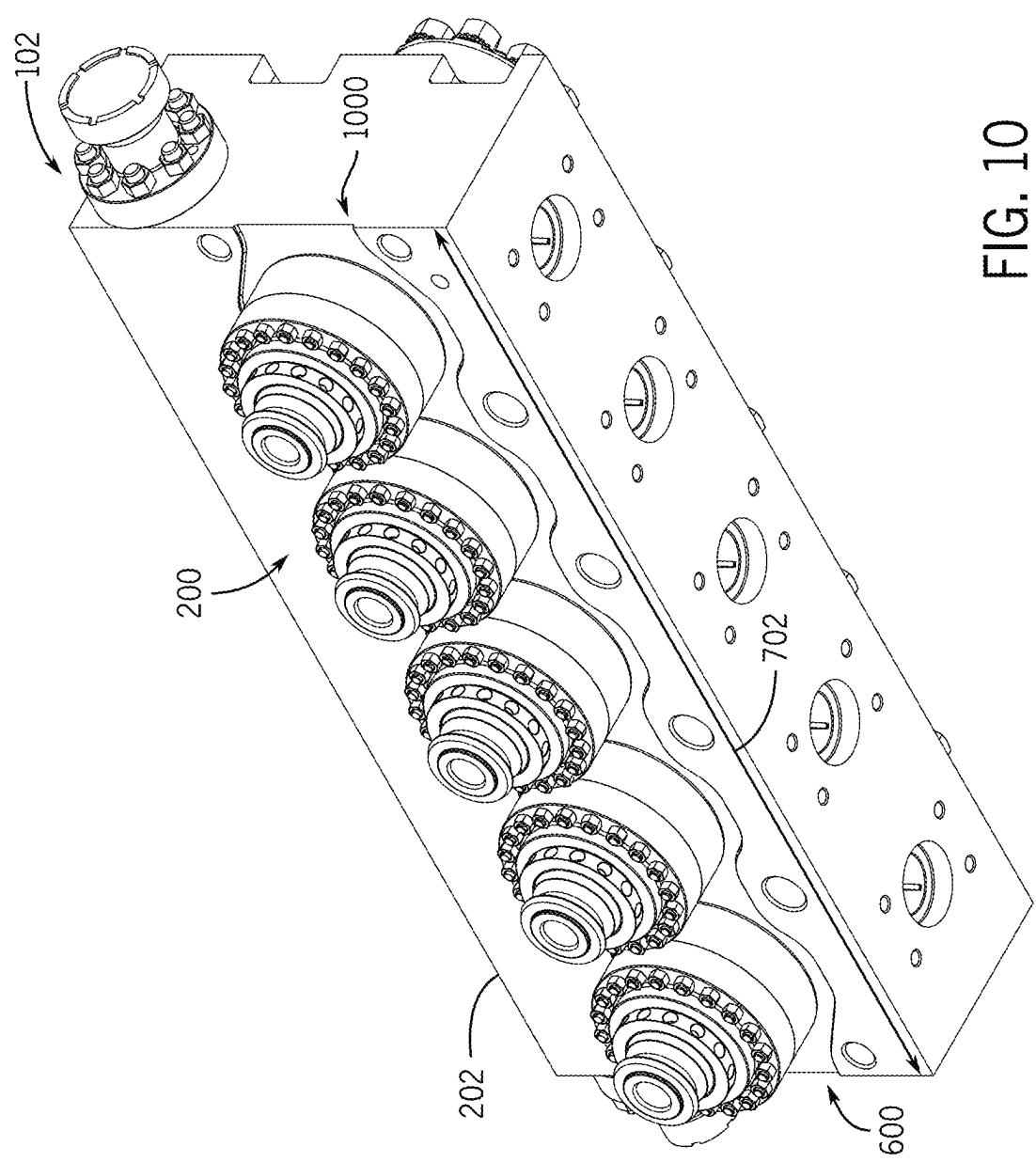


FIG. 10

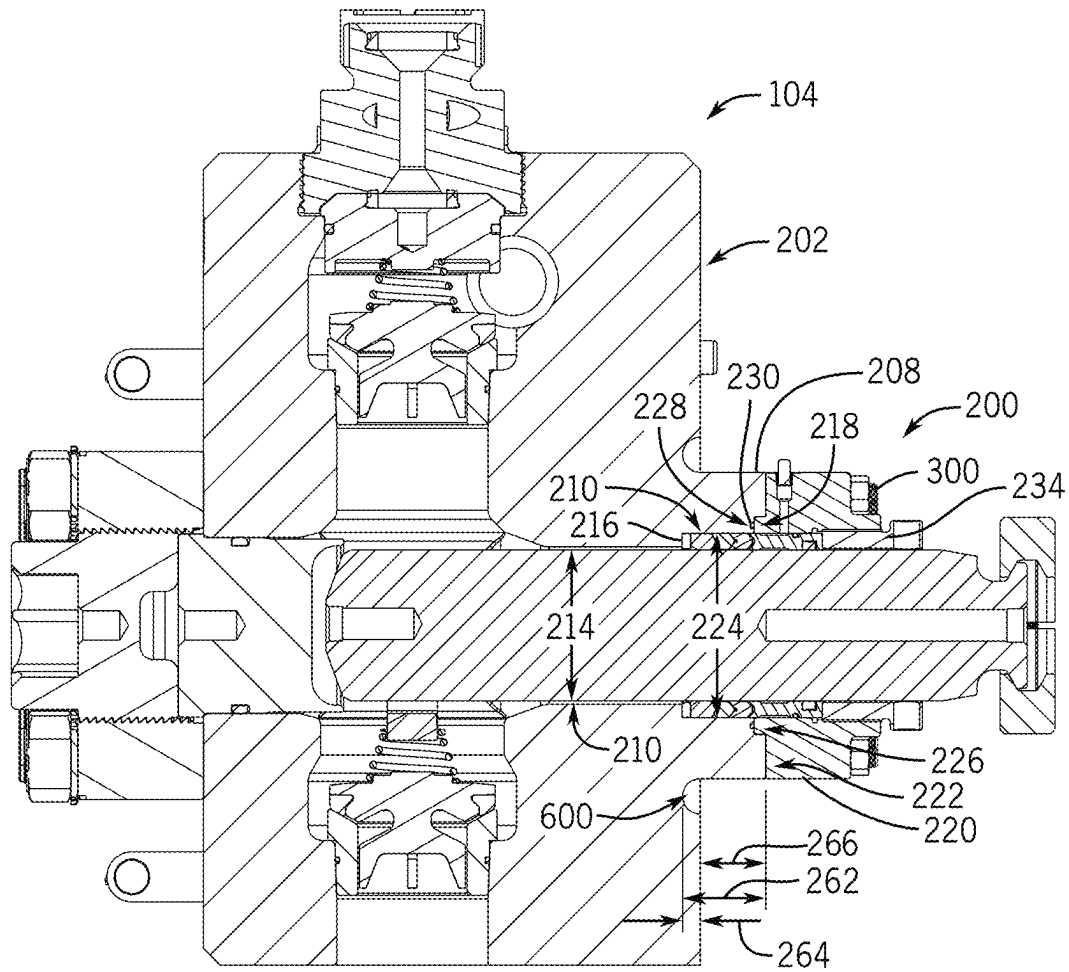


FIG. 11

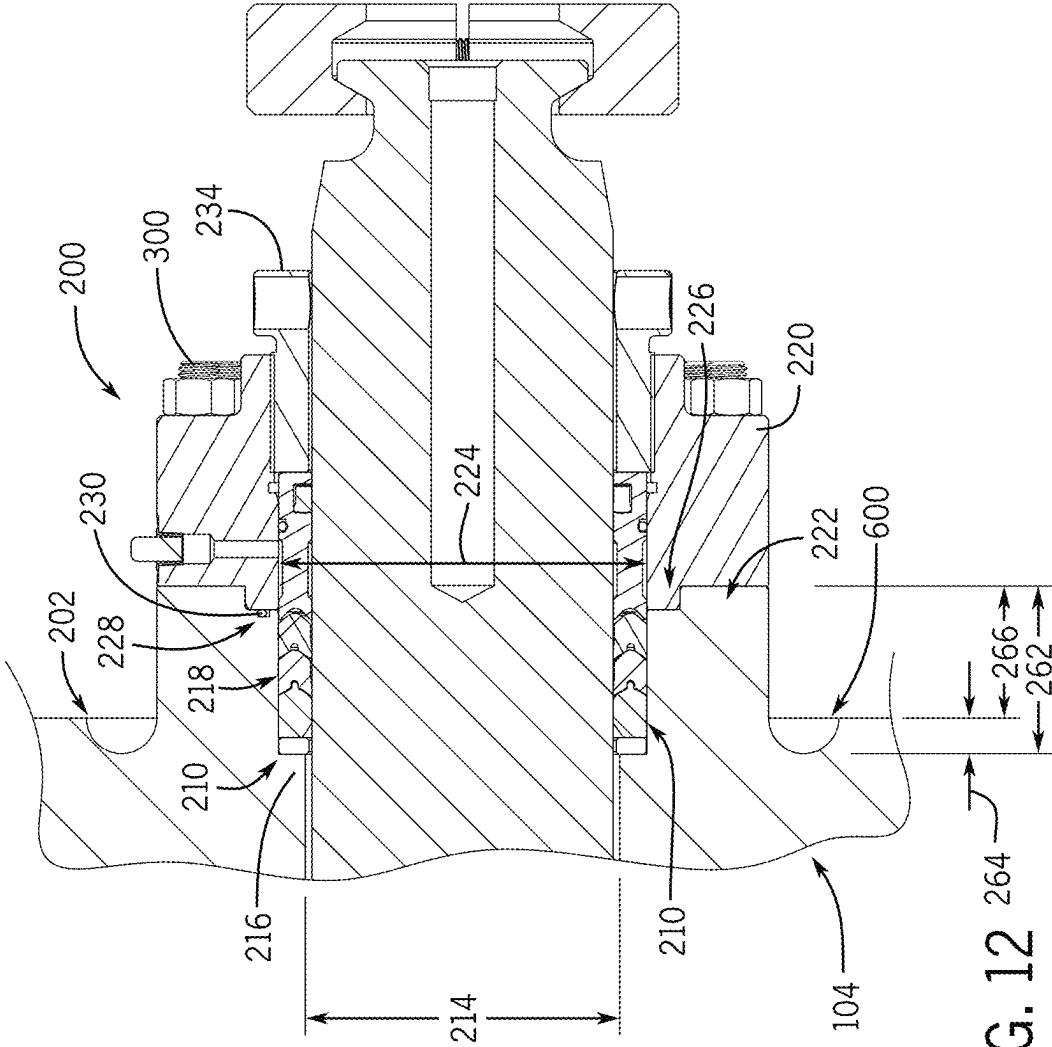


FIG. 12

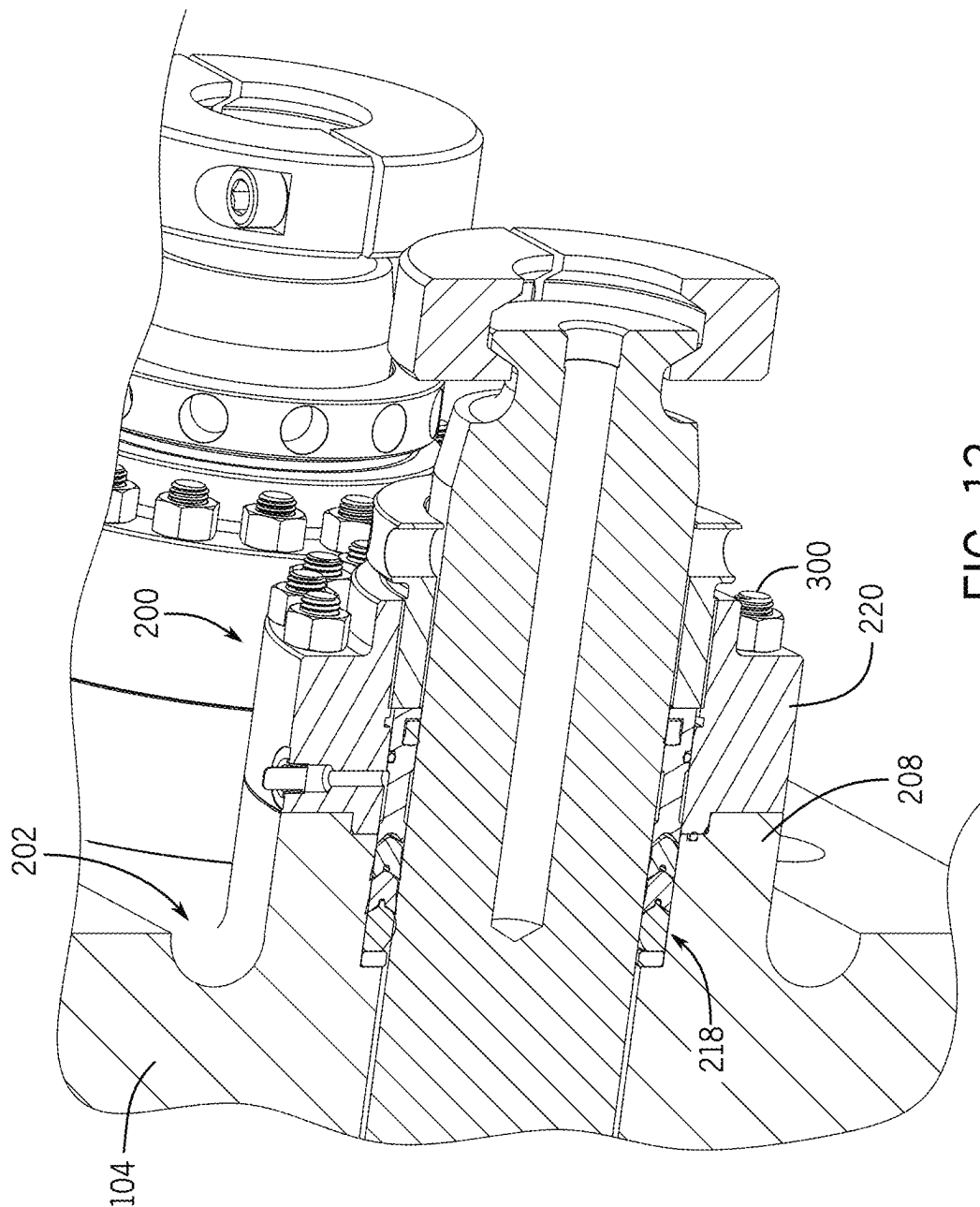


FIG. 13

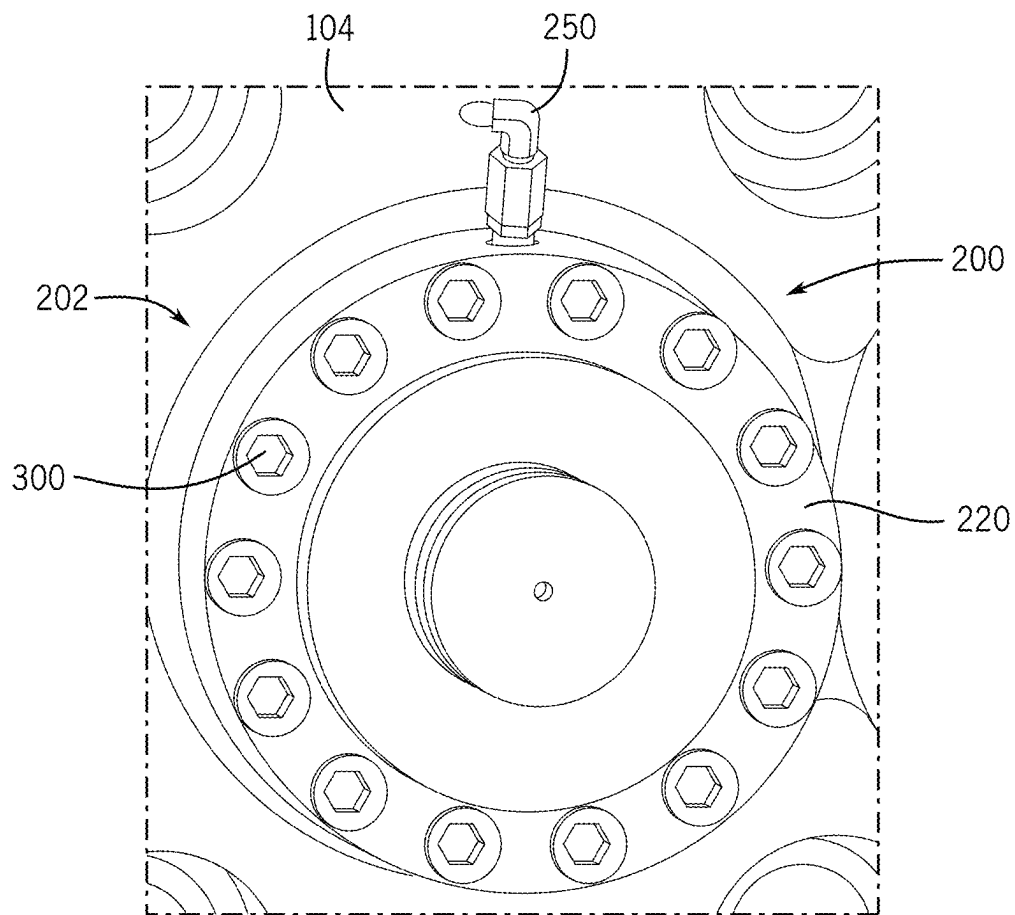


FIG. 14

1

DUAL RING STUFFING BOX**CROSS REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. Non-Provisional application Ser. No. 17/864,873, filed Jul. 14, 2022, titled “DUAL RING STUFFING BOX,” which is a continuation-in-part of U.S. patent application Ser. No. 17/213,854, filed Mar. 26, 2021, titled “DUAL RING STUFFING BOX,” now U.S. Pat. No. 11,391,374, issued Jul. 19, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/137,501, filed Jan. 14, 2021, titled “DUAL RING STUFFING BOX,” and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,314, filed Feb. 4, 2021, titled “FLUID END FOR A PUMPING SYSTEM,” now U.S. Patent No. D933104, issued Oct. 12, 2021, and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,318, filed on Feb. 4, 2021, titled “FLUID END FOR A PUMPING SYSTEM,” now U.S. Patent No. D933105, issued Oct. 12, 2021, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

TECHNICAL FIELD

Embodiments of the subject matter disclosed herein generally relate to pump systems, and in particular to sealing assemblies for pumping systems.

BACKGROUND

Pumping systems may be used in a variety of applications, such as industrial applications where pumping systems are used to elevate a working fluid pressure. One such application is hydraulic fracturing systems, where high pressure pumps are used to increase a fluid pressure of a working fluid (e.g., fracturing fluid, slurry, etc.) for injection into an underground formation. The working fluid may include particulates, which are injected into fissures of the formation. When the fluid is removed from the formation, the particulates remain and “prop” open the fissures, facilitating flow of oil and gas. In many applications, reciprocating pumps are used where a fluid is introduced into a fluid end inlet passage and out through an outlet passage. A plunger reciprocates within a bore to add energy to the fluid. A stuffing box may be positioned around the plunger to block fluid leaks.

SUMMARY

Applicant recognized the problems noted above herein and conceived and developed embodiments of systems and methods, according to the present disclosure, for stuffing boxes.

In an embodiment, a system includes a pressure ring positioned to abut a fluid end face, the pressure ring having a pressure ring bore with a diameter larger than a fluid end bore diameter. The system also includes an adjacent ring coupled to a fluid end, the pressure ring positioned between the adjacent ring and the fluid end face. The system further includes a face seal positioned at an interface of the pressure ring and the fluid end face, the face seal containing pressurized fluid associated with the fluid end. The system also includes packing material positioned within at least a portion of the pressure ring bore.

2

In an embodiment, a system includes a fluid end, the fluid end having a recessed area extending to a fluid end face. The system further includes a pressure ring positioned within the recessed area against the fluid face to compress the face seal, the pressure ring having a pressure ring bore positioned co-axially with a fluid end bore, and the pressure ring bore has a diameter greater than the fluid end bore. The system also includes an adjacent ring positioned against the pressure ring and coupled to the fluid end. The system includes a face seal arranged between the pressure ring and the fluid end. The system further includes packing material positioned within the pressure ring bore, at least a portion of the packing material being positioned external to the fluid end bore.

In an embodiment, a method includes inserting a face seal into a face groove. The method also includes positioning a pressure ring, within a recessed portion, to compress the face seal. The method further includes positioning an adjacent ring against the pressure ring. The method also includes securing the adjacent ring to a fluid face, wherein the pressure ring is positioned between the adjacent ring and the fluid end. The method includes installing, within a pressure ring bore, packing material.

In an embodiment, a system includes a fluid end having a fluid end face and a recess extending toward a base. The system also includes a pressure ring extending from the fluid end face, the pressure ring having a pressure ring bore diameter larger than a fluid end bore diameter. The system further includes an adjacent ring coupled to the fluid end, the pressure ring positioned between the adjacent ring and the base, wherein at least one fastener extends through an aperture of the adjacent ring to couple the adjacent ring to the fluid end. The system includes packing material positioned within at least a portion of the pressure ring bore, the packing material positioned such that axial movement toward the fluid end is blocked by the base.

In an embodiment, a system includes a fluid end, the fluid end having a recessed area. The system also includes a pressure ring extending from the recessed area, the pressure ring having a pressure ring bore positioned co-axially with a fluid end bore, and the pressure ring bore having a diameter greater than the fluid end bore. The system further includes an adjacent ring positioned against the pressure ring and coupled to the fluid end. The system includes packing material positioned within the pressure ring bore, at least a portion of the packing material being positioned external to the fluid end bore. The system also includes an injection port extending through the adjacent ring, the injection port arranged in a low pressure region, axially away from a seal between the pressure ring and the adjacent ring.

In an embodiment, a method includes providing a fluid end including a recessed area extending into a fluid end face and a pressure ring extending axially out of the recessed area. The method also includes positioning an adjacent ring against the pressure ring. The method further includes securing the adjacent ring to the fluid end. The method includes installing, within a pressure ring bore, packing material.

BRIEF DESCRIPTION OF THE DRAWINGS

The present technology will be better understood on reading the following detailed description of non-limiting embodiments thereof, and on examining the accompanying drawings, in which:

FIG. 1 is a schematic cross-sectional view of a prior art pump assembly;

3

FIG. 2 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 3 is an isometric view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 4 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 5 is an isometric view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 6 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 7 is an isometric view of an embodiment of a fluid end, in accordance with embodiments of the present disclosure;

FIG. 8 is an isometric view of an embodiment of a fluid end, in accordance with embodiments of the present disclosure;

FIG. 9 is an isometric view of an embodiment of a fluid end with a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 10 is an isometric view of an embodiment of a fluid end with a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 11 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 12 is a cross-sectional view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure;

FIG. 13 is a sectional isometric view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure; and

FIG. 14 is a perspective view of an embodiment of a stuffing box assembly, in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

The foregoing aspects, features, and advantages of the present disclosure will be further appreciated when considered with reference to the following description of embodiments and accompanying drawings. In describing the embodiments of the disclosure illustrated in the appended drawings, specific terminology will be used for the sake of clarity. However, the disclosure is not intended to be limited to the specific terms used, and it is to be understood that each specific term includes equivalents that operate in a similar manner to accomplish a similar purpose.

When introducing elements of various embodiments of the present disclosure, the articles “a”, “an”, “the”, and “said” are intended to mean that there are one or more of the elements. The terms “comprising”, “including”, and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Any examples of operating parameters and/or environmental conditions are not exclusive of other parameters/conditions of the disclosed embodiments. Additionally, it should be understood that references to “one embodiment”, “an embodiment”, “certain embodiments”, or “other embodiments” of the present disclosure are not intended to be interpreted as excluding the existence of additional embodiments that also incorporate the recited features. Furthermore, reference to terms such as “above”, “below”, “upper”,

4

“lower”, “side”, “front”, “back”, or other terms regarding orientation or direction are made with reference to the illustrated embodiments and are not intended to be limiting or exclude other orientations or directions. Moreover, like numbers for may be used for like components, but such disclosure is for convenience purposes and is not intended to limit the scope of the present disclosure.

Traditional fluid ends feature an integrated stuffing box which requires a larger forging and more material to be removed. This is an expensive process. An alternative is a removable and replaceable stuffing box where the pressure vessel components are attached to the fluid end. This has been achieved previously by stuffing boxes which were threaded into the fluid end via large power threads such as ACME or Buttress profiles. These threaded stuffing boxes were heavy and difficult to install and then prone to failures of the threaded interfaces.

Embodiments of the present disclosure address and overcome problems faced with traditional stuffing boxes utilized with pumping units, such as fluid ends for fracturing pumps, and may be further incorporated in various systems with reciprocating components. In at least one embodiment, systems and methods of the present disclosure include a set of rings flanged into place such that a pressure containing ring is made from a wear resistant stainless steel, or any other type of wear resistant material, and an adjacent ring is made from a less expensive alloy steel, or any type of compatible material, as well as similar wear resistant materials. The two rings are joined together and preloaded by a series of fasteners (e.g., studs, nuts, bolts, cap screws, etc.). In various embodiments, the pressure containing ring houses a face seal or the fluid cylinder may house the face seal. Furthermore, the seal could be partially set into both. In one or more embodiments, a sealing element is arranged between the two rings, which includes a sealing component, such as an o-ring or other seal, which may be retained in the pressure ring, the adjacent ring, or both. Various embodiments also include a cross drilled weep hole where the intersecting holes may or may not be drilled entirely through the pressure ring. For example, in some iterations they may be intersecting blind holes. The pressure ring extends past the last pressure ring of a packing set such that anything closer to the exterior of the fluid end (e.g., radially outward from a pressure chamber) is not exposed to pressurized fracking fluid.

In one or more embodiments, systems and methods use studs and nuts that pass through dual rings of the stuffing box. It should be appreciated that these fasteners are provided by way of example only and are not intended to limit the scope of the present disclosure, as other fasteners, such as bolts and the like, may be used. Moreover, a threaded or otherwise locking interface between the rings may be utilized, such as a bayonet connection, J-hook, or the like. The fluid end may be machined such that an outside diameter of the pressure ring serves as a fine alignment feature during installation. The pressure ring may also be arranged such that it is also the only portion of the stuffing box exposed to the high pressure fracking fluid. Accordingly, in various embodiments, the pressure ring may be formed from a high grade stainless steel, among other wear resistant and/or specialized materials, while the adjacent ring would be made from alloy steel, among other types of materials, thereby reducing costs associated with implementation of the various embodiments disclosed herein. In at least one embodiment, the pressure ring may be integrally formed or be continuous with the materials associated with the fluid end.

5

Embodiments of the present disclosure may incorporate a sealing element within the fluid end body and/or within the pressure ring. This configuration may further include a single outer diameter. As a result, a seal designed to seal against the pressurized fluid may be retained by the pressure ring and/or the adjacent ring. Furthermore, embodiments may include a grease insertion point associated with the adjacent ring and also one or more threads to receive a packing nut. Moreover, a recessed shelf may be incorporated such that radial holes in the packing nut are clearly exposed above the heads of the fasteners (e.g., studs and nuts, bolts, cap screws, etc.). In certain embodiments, the packing set rests on the fluid end block surface. In others, the pressure containing ring features a shelf where a spacer ring may be installed and retained such that the packing assembly sits on the spacer ring.

In one or more embodiments, a section of removed material across a backside of the fluid end, such that the stuffing box does not sit within any bore but sits on the back face of the fluid end, is incorporated. For example, a recess or groove may be formed within the fluid end to receive the pressure ring. In one or more embodiments, the recess or groove may extend substantially across the entire fluid end. In certain embodiments, the recess or groove may be particularly located at individual locations for individual stuffing boxes. Furthermore, the pressure ring may extend from the fluid end face such that the groove or recess extends radially into the body of the fluid end. As will be described below, in various embodiments, the recess or groove may be squared or slanted. Furthermore, a depth of the recess or groove may be particularly selected based on one or more operating conditions for the fluid end.

FIG. 1 is a schematic cross-sectional view of an embodiment of a pump assembly 100, which may also be referred to as a reciprocating pump assembly and/or a reciprocating pump. The pump assembly 100 may be utilized during hydraulic fracturing operations, among other operations, where a working fluid (e.g., fracturing fluid, slurry, etc.) is introduced into the pump and energy is added to the working fluid to increase a pressure of the working fluid. Fracturing fluid, by way of example only, may include corrosives and also particulates, such as sand or ceramics, which are utilized during fracturing operations. These corrosives and particulates cause erosion within the pump assembly 100, which may undesirably affect fracturing operations and lead to down times to replace various components. Additionally, the fracturing fluids may include corrosive acids and the like, which may wear down components of the pump assembly 100.

It should be appreciated that various components of the pump assembly 100 have been removed for clarity with the following discussion. For example, a power end has been removed in favor of focusing on the illustrated fluid end 102 of the pump assembly 100. The power end may include a crankshaft that is driven by an engine or motor to facilitate operations. The fluid end 102 includes a fluid end block 104 that may house one or more components discussed herein. A plunger rod 106 is driven (e.g., via the crankshaft) to reciprocate within the fluid end block 104 along a plunger axis 108. The plunger rod 106 is positioned within a bore 110 extending through at least a portion of the fluid end block 104. The illustrated bore 110 is arranged along the plunger axis 108 (e.g., first axis) and intersects a pressure chamber 112, which is arranged along a pressure chamber axis 114 (e.g., second axis), which is positioned substantially perpendicular to the plunger axis 108. It should be appreciated that the pump assembly 100 may include multiple

6

plunger rod and pressure chamber arrangements, which may be referred to as a plunger throw. For example, the pump assembly 100 may be a triplex pump, quadplex pump, quintuplex pump, and the like.

The illustrated fluid end block 104 includes an inlet passage 116 and an outlet chamber 118, which are generally coaxial and arranged along the pressure chamber axis 114. In other words, the inlet passage 116 and the outlet chamber 118 are axially aligned with respect to one another and/or the pressure chamber 112. In various embodiments, fluid enters the pressure chamber 112 via the inlet passage 116, for example on an up stroke of the plunger rod 106, and is driven out of an outlet passage 120, for example on a down stroke of the plunger 106.

Respective valve assemblies 122, 124 are arranged within the inlet passage 116 and the outlet chamber 118. These valve assemblies 122, 124 are spring loaded in the illustrated embodiment, but it should be appreciated that such an arrangement is for illustrative purposes only. In operation, a differential pressure may drive movement of the valve assemblies. For example, as the plunger rod 106 is on the upstroke, pressure at the inlet passage 116 may overcome the spring force of the valve assembly 122, thereby driving fluid into the pressure chamber 112. However, on the down stroke, the valve assembly 122 may be driven to a closed position, while the spring force of the valve assembly 124 is overcome, thereby enabling the fluid to exit via the outlet passage 120.

Piston pumps or plunger pumps, such as the pump assembly 100 shown in FIG. 1, are positive displacement pumps and are commonly used in environments where the fluids that are being handled pose problems such as high temperatures, viscous media, or solids-charged liquids. One such example is in oil and gas operations, particularly fracturing operations, where solids laden fluids may be used. Examples of these fluids include drilling fluids, muds, cement slurries, fracturing slurries, acids and the like, which frequently must be pumped under high pressure into the well. These abrasive fluids provide challenges for the various sealing interfaces of the pumps. One such interface includes along the plunger 106. As will be described, various systems and methods of the present disclosure may include a stuffing box configuration that includes one or more rings to secure a packing material along the bore 110 to block leakage of fluid out of the fluid end 102. In various embodiments, the one or more rings may be arranged external to the bore 110, such as against a face of the fluid end 102.

FIG. 2 is a cross-sectional side view of an embodiment of a stuffing box assembly 200 that may be utilized with one or more reciprocating components, such as the plunger 106 illustrated in FIG. 1. As noted, like numerals may be used throughout to represent like features, but such numbering for convenience and is not intended to limit the scope of the present disclosure. In this example, the stuffing box assembly 200 is positioned on a back side 202 of the fluid end 102. It should be appreciated that “back side” is a relative term and is not intended to limit one or more orientations of the fluid end 102. The plunger 106 extends within the bore 110 along the plunger axis 108, as indicated above, and may be driven to reciprocate along the plunger axis 108 to add energy to the pressure chamber 112 (FIG. 1).

The illustrated stuffing box assembly 200 is positioned against a recessed area 204 (e.g., grooved area, offset area, depressed area, etc.) formed in the back side 202. In at least one embodiment, at least a portion of the stuffing box assembly 200 may extend at least partially into a recessed portion or cut out section formed with respect to the recessed

area **204**. In other words, the recessed area **204** represents a removal of material from the fluid end **102** and/or a forming process that provides an area with a reduced lateral position (with respect to the orientation of FIG. 2) compared to an adjacent area. In this example, the recessed area **204** includes a wall **206** forming the recessed area **204**, which may be a continuous, annular wall **206**. It should be appreciated that reference to a single wall is not intended to be limiting, as there may be various regions formed that have different radial extents or the like to prevent rotation and, moreover, the wall **206** may not be continuous in all embodiments. As an example, in one or more embodiments, the wall **206** extends for a length of the fluid end **102**. However, in other examples, the wall **206** may extend for a particular extent. In one or more embodiments, the recessed portion and/or groove **206** may be omitted and the stuffing box assembly **200** may be pressed directly against the back side **202**, for example along a planar region of the back side **202**. By way of example only, a number of different grooves **204** may be formed along the back side **202**, for example at each bore **110**. In this example, the recessed area **204** receives a pressure ring **208** (e.g., a first ring, a pressure-containing ring) that abuts to the recessed area **204** such that at least a portion of the pressure ring **208** is overlapped by at least a portion of the wall **206**. That is, the pressure ring **208** abuts against the recessed area **204** and, in various embodiments, at least a portion of the pressure ring **208** extends axially out of recessed portion associated with the recessed area **204** (e.g., axially along the plunger **108** in a direction away from the back side **202**). It should be appreciated that a distance of axial extension may be particularly selected based on design conditions, and moreover, that a recessed area depth may also be particularly selected based on design conditions. As noted, in various embodiments, the recessed area **204** may be omitted such that the stuffing box assembly **200** is flush against the back side **202**. The illustrated pressure ring **208** further includes a pressure ring bore **210** that is axially aligned with the bore **110** to enable reciprocation of the plunger **106** along the plunger axis **108**. However, as will be described below, various dimensions of the pressure ring bore **210** may differ from the bore **110** to facilitate installation of various materials, such as one or more packing assemblies or components thereof.

In the example of FIG. 2, an inner diameter **212** of the pressure ring **208** is greater than an inner diameter **214** of the bore **110**, thereby providing a shelf **216** for packing materials **218** (e.g., stuffing material, a packing assembly). The packing materials **218** include various components, such as seals, brass, and the like, which may be utilized to block fluid flow upward and along the plunger rod **106**. The packing materials **218** may be particularly selected based, at least in part, on one or more operating conditions of the fluid end **102**, as well as various dimensions of the shelf **216**, plunger **106**, and pressure ring **208**, among other components. Accordingly, the illustrated configuration is for example purposes only, and it should be appreciated that more or less material may be included in embodiment where additional or less space is provided for the packing materials **218**. It should further be appreciated that, in various embodiments, the shelf **216** may be formed, at least in part, by the pressure ring **208**. For example, the inner diameter **212** may not be constant across the pressure ring **208** and may include the shelf **216** at a variable diameter portion. That is, the pressure ring **208** may include a first inner diameter at the inner diameter **212** and a second inner diameter, less than the first inner diameter **212**, that forms the shelf. However, in various other embodiments, the first diameter **212** (e.g., the

portion that engages the packing materials **218**) may be smaller than a second inner diameter. Furthermore, one or more components, such as a spacer ring, may also be incorporated to serve as the shelf **216**.

In various embodiments, an adjacent ring **220** (e.g., outer ring, second ring, non-pressure containing ring, etc.) is positioned to abut the pressure ring **208**. In this example, the adjacent ring **220** is at least partially overlapped by the pressure ring **208**. That is, an overlapped region **222** includes a second pressure ring inner diameter **224**, larger than the inner diameter **212**, that receives a lip **226** of the adjacent ring **220**. As a result, a portion of the pressure ring **208** is illustrated as radially overlapping a portion of the adjacent ring **220**. In various embodiments, the lip **226** is proximate a seal groove **228** formed in the pressure ring **208** that receives a seal **230**. It should be appreciated that the seal groove **228** and the position of the seal **230** may also be within the adjacent ring **220** in other embodiments or may be partially within the pressure ring **208** and partially within the adjacent ring **220**.

In various embodiments, the overlapped region **222** is configured to be axially farther from the back side **202** (e.g., toward the right relative to the orientation of FIG. 2) with respect to one or more packing materials **218** such that the adjacent ring **220** is not exposed to pressurized fluid from the fluid end **102**. It should be appreciated that the overlapped region **222** may also be positioned along any portion of the packing materials **218** and that the configuration in FIG. 2 is for illustrative purposes only.

The adjacent ring **220** further includes a coupling region **232**, which includes threads in this embodiment, to receive a retainer **234** (e.g., packing nut), which may include mating threads or the like. The retainer **234** may be utilized to maintain a position of the packing materials **218**. It should be appreciated that the retainer **234** may be coupled to the adjacent ring **220** and/or to other components in a variety of ways, such as external fittings, clamps, and the like. As shown, the coupling region **232** is arranged at a first end **236** of the adjacent ring **220**, opposite a second end **238**. The first end **236** is farther from the back side **202** than the second end **238**. The first end **236** further includes a stepped portion **240** such that a stepped portion length **242** is less than an adjacent ring length **244**. The stepped portion **240** provides a region to receive one or more fasteners to couple the adjacent ring **220** to the pressure ring **208**. For example, fasteners may extend through aligned apertures to join the components together. As illustrated, the stepped portion **240** positions the fasteners out of alignment with the retainer **234** to provide improved access to both components. In other words, the fasteners are arranged axially closer to the back side **202** than a top of the retainer **234** to facilitate access to each of the components. It should be appreciated that the fasteners illustrated are for illustrative purposes only and that other coupling devices may be utilized to join the pressure ring **208** to the adjacent ring **220**.

Returning to the pressure ring **208**, a weep hole **246** is illustrated extending radially through the pressure ring **208**. In this example, the weep hole **246** intersects an axial hole **248** extending axially through the pressure ring **208**. It should be appreciated that one or more of the weep hole **246** or the axial hole **248** may or may not extend entirely through the pressure ring **208**. For example, one or more of the holes **246**, **248** may extend to a particular region to form a blind. In this example, the weep hole **246** is positioned proximate the overlapping region **222**, but various other locations may also include weep holes **246**. That is, the weep holes **246** may be arranged at different locations along the axial length

of the pressure ring 208. Furthermore, one or more weep holes 246 may be drilled circumferentially about the pressure ring 208 that intersect with one or more axial holes 248 or that do not intersect axial holes 248. By way of example, the pressure ring 208 may include multiple weep holes 246 arranged at different circumferential positions of the pressure ring 208 and one or more of these weep holes 246 may or may not intersect one or more additional axial holes 248.

The illustrated adjacent ring 220 also includes an injection port 250, which may be utilized to direct grease toward the packing material 218. It should be appreciated that the injection port may be arranged at any reasonable location to enable injection of various substances toward the bore 110. In various embodiments, it may be preferential to position the injection port 250 downstream of the seal 230 such that the injection port 250 is not subject to the pressure of the fracturing fluid.

In operation, the pressure ring 208 and the adjacent ring 220 are coupled to the fluid end 102, for example via fasteners. As illustrated, the back side 202 may include a face seal 252 between the back side 202 and the pressure ring 208. In this example, a face groove 254 is formed in the back side 202 to receive the face seal 252. It should be appreciated that, in other embodiments, the face seal 252 may be associated with the pressure ring 208, rather than the back side 202. For example, the face groove 254 may be formed in the pressure ring 208. Additionally, in at least one embodiment, the face groove 254 may be formed partially within the back side 202 and partially within the pressure ring 208. In one or more embodiments, the face seal 252 may seal against pressurized fluid within the fluid end 102. As shown, embodiments provide the packing materials 218 of the associated stuffing box assembly 200 abutting on the fluid end 102, rather than a sleeve being within the bore, as noted above.

Various embodiments of the present disclosure are directed toward a substantially external stuffing box assembly 200 that is not positioned within the bore 110, however, it should be appreciated that various alternative configurations may include the stuff box assembly 200 or portions thereof that extend into the bore 110. In this example, the groove 204 is formed within the back side 202 such that the groove 204 extends a groove depth 256 into the back side 202. The groove depth 256, as noted above, may be particularly selected based on operating conditions, tooling conditions, or the like. The illustrated groove 204 is formed with a groove height 258 that is configured to receive the pressure ring 208 with a secure fit such that the groove height 258 may be substantially equal to a pressure ring outer diameter 260. In embodiments, the configuration of the dimensions may facilitate alignment and installation of the components, where a misalignment may be visually established by resistance of fitting the pressure ring 208 into the groove 204. Other embodiments may include configurations where the groove height 258 is greater than the outer diameter 260.

As described above, various embodiments position the stuffing box assembly 200 external to the bore 110. In the illustrated embodiment, the pressure ring 208 extends axially away from the back side 202 such that at least a portion of a pressure ring thickness 262 is external the groove 204. That is, the pressure ring thickness 262 may include an internal portion 264 that is within the groove 204 and an external portion 266 that is outside of the groove 204. In various embodiments, the internal portion 264 is less than the external portion 266, but it should be appreciated that a variety of configurations may be utilized within the scope of

the present disclosure (e.g., internal greater than external, portions equal, etc.). Accordingly, the pressure ring 208 may be fixed to the fluid end 102 via fasteners 268 that extend through the pressure ring 208 and the adjacent ring 220. In one or more embodiments, the fasteners 268 include threaded fasteners, such as bolts, but it should be appreciated that additional fasteners may also be used within the scope of the present disclosure, such as clamps, radial fasteners (as opposed to the illustrated axial fasteners), integrated bayonet fittings, and the like.

In various embodiments, the back side 202 may also be referred to as having a fluid end face 270, which may correspond to the surface within the groove 204. As a result, in the illustrated example, the pressure ring 208 would abut or otherwise be positioned against the fluid end face 270. Moreover, an outer face 272 may be used to describe the region around the groove 204, where the fluid end face 270 is axially closer to the internal components of the fluid end 102 than the outer face 272.

FIG. 3 is an isometric view of an embodiment of the stuffing box assembly 200. In this example, the pressure ring 208 and the adjacent ring 220 are both illustrated extending axially away from the back side 202. Further illustrated are the fasteners 268 that extend through the pressure ring 208 and the adjacent ring 220 to couple the stuffing box assembly 200 to the back side 202. For example, the fasteners 268 may extend into mating apertures formed in the fluid end 102. Moreover, as noted above, the retainer 234 is arranged within the adjacent ring 220, for example coupled via threads. The injection port 250 is arranged positioned proximate the weep whole 246. As noted above, while a single port 250 and single weep hole 246 are illustrated in this example, more may be included, for example at different circumferential positions.

FIG. 4 is a cross-sectional view of an embodiment of a stuffing box assembly 400 that may share one or more features with the stuffing box assembly 200 of FIG. 2. For simplicity and clarity, certain portions will not be repeated herein. In this example, the pressure ring 208 is arranged within the recessed area 204 and abuts the back side 202 of the fluid end 102. As described above, the internal portion 264 extends into the groove depth 256. In this example, the internal portion 264 is less than the external portion 266. In at least one embodiment, the pressure ring 208 compresses the face seal 252, which in this example is shown associated with the back side 202 within the face groove 254, but in other examples may be associated with the pressure ring 208. Furthermore, as described above, the adjacent ring 220 is coupled to the pressure ring 208.

In this example, a spacer ring 402 is positioned along a transition 404 at the bore 110. As shown, the spacer ring 402 is arranged proximate the face seal 252 and extends axially away from the back side 202 such that the spacer ring 402 at least partially overlaps the face groove 254. It should be appreciated that, in various embodiments, a conjoined groove or set of grooves may be utilized to receive the face seal 252 and the spacer ring 402. Moreover, in one or more embodiments, material may separate grooves for the face seal 242 and the spacer ring 402. In various embodiments, the spacer ring 402 may even form a portion of the bore 110. The spacer ring 402 may be utilized as the shelf 216, as described above, to receive the packing materials 218.

As shown, the pressure ring 208 may include a notch 406 that receives a portion of the spacer ring 402. For example, the notch 406 may represent a groove or an increased diameter portion 408, compared to the inner diameter 212, to receive and support at least a portion of the spacer ring

11

402. In various embodiments, the pressure ring 208, in combination with one or more other elements, at least partially blocks axial movement of the spacer ring 402 away from the back side 202 (toward the right relative to the orientation of FIG. 4). It should be appreciated that relative dimensions for the spacer ring 402 and/or notch 406 may be particularly selected based on operating conditions. In operation, the illustrated stuffing box assembly 400 enables isolation of the pressurized fluids within the fluid end 102 by supporting one or more components against the back side 202 of the fluid end 102.

As noted above, in this example the pressure ring 208 includes three different internal diameter positions, namely the inner diameter 212, the second inner diameter 224, and the increased diameter portion 408. In this example, the increased inner diameter portion 408 is smaller than the second inner diameter 224, but larger than the inner diameter 212. Accordingly, it should be appreciated that various different stepped portions may be positioned along the pressure ring bore 210 to facilitate positioning and retention of one or more components utilized with the stuffing box assembly 200.

FIG. 5 is an isometric view of an embodiment of the stuffing box assembly 400. In this example, the pressure ring 208 and the adjacent ring 220 are both illustrated extending axially away from the back side 202. Further illustrated are the fasteners 268 extending through the pressure ring 208 and the adjacent ring 220 to couple the stuffing box assembly 400 to the back side 202. For example, the fasteners 268 may extend into mating apertures formed in the fluid end 102. Moreover, as noted above, the retainer 234 is arranged within the adjacent ring 220, for example coupled via threads. The injection port 250 is arranged positioned proximate the weep hole 246. As noted above, while a single port 250 and single weep hole 246 are illustrated in this example, more may be included, for example at different circumferential positions.

FIG. 6 is a cross-sectional view of an embodiment of the stuffing box assembly 200 coupled to a back side 202 of the fluid end 102. In this example, the recessed area 204 is represented by a cut out region 600, which will be described in more detail below. In this example, the cut out region 600 includes a sloped entry 602 and a mating face 604, which represents at least a portion of the back side 202 of the fluid end 102. As shown, the cut out region 600 includes a depth 606 (similar to the groove depth 256), which may be representative of removed material or the part may be cast or otherwise formed without this material. As a result, inclusion of additional components abutting the back side 202 may not substantially increase an overall size of the fluid end 102. While the illustrated embodiment includes the recessed area 204 as the cut out region 600, it should be appreciated that various embodiments may include an extruded or outwardly extended region as opposed to a cut out region. By way of example, the portions illustrated as the cut out region 600 may extend outwardly and away such from back side 202 as opposed as being recessed. Moreover, as will be described below, additional geometries may also be utilized to form the cut out region 600 and/or any extruded or outwardly extending region.

It should be appreciated that the sloped entry 602 is for illustrative purposes only, along with the substantially symmetrical configuration of the cut out region 600. For example, the sloped entry 602 may be replaced by squared (e.g., approximately 90 degree) walls and/or a curved entry. Furthermore, in various embodiments, the cut out region 600 extends along a length of the fluid end, as illustrated in

12

below. However, in other embodiments, individual cut out regions 600 may be formed from the fluid end. In operation, the stuffing box assembly 200 may couple to the back end 202, within the cut out region 600, to compress the face seal 252, as described above.

FIG. 7 is a schematic view of an embodiment of the fluid end 102 illustrating the cut out region 600. In this example, the cut out region 600 is substantially squared. That is, the sloped entry 602 (FIG. 6) is replaced by a squared entry 700, which may be substantially 90 degrees. Like the sloped entry 602, the squared entry 700 may be referred to as having a polygonal profile. The squared entry 700 transitions to the depth 606, as noted above. Moreover, the cut out region 600 extends for a fluid end length 702, but it should be appreciated that the cut out region 600 may be segmented or not extend the full length 702. The illustrated depth 606 is also particularly selected based on operating conditions.

In one or more embodiments, the fluid end 102 is forged or otherwise manufactured in the configuration shown, which may result in less material being utilized in operations. In one or more embodiments, the fluid end 102 may also be machined to include the cut out region 600. Accordingly, the general shape of the cut out region 600 may also be selected, at least in part, with manufacturing considerations in mind.

FIG. 8 is a schematic view of an embodiment of the fluid end 102 illustrating the cut out region 600. In this example, the sloped entry 602 is visible, as shown in FIG. 6. Moreover, the cut out region 600 extends for a fluid end length 702, but it should be appreciated that the cut out region 600 may be segmented or not extend the full length 702. The illustrated depth 606 is also particularly selected based on operating conditions. This example further includes the face groove 254 for the face seal 252 (FIG. 2) and the apertures for receiving the fasteners 268 (FIG. 2).

FIG. 9 is an isometric view of an embodiment of the fluid end 102 including the stuffing box assemblies 200 coupled to the back side 202 within the cut out region 600. In this example, the cut out region 600 extends for the fluid end length 702. As noted above, in various embodiments of the present disclosure, stuffing box assemblies 200 are arranged such that one or more components bear against the back side 202 of the fluid end 102 and without extending into the bore 100.

It should be appreciated that various embodiments of the present disclosure may incorporate materials formed from similar or dissimilar materials. By way of example, the pressure ring 208 may be formed from a stronger, more resistant material than the adjacent ring 220 due to the pressure ring 208 being in contact with pressurized fluid while the adjacent ring 220 may be isolated from the fluid. Accordingly, less expensive materials may be utilized for the adjacent ring 220. It should be appreciated that all materials used may be particularly selected for expected operating conditions.

FIG. 10 is a bottom perspective view of an embodiment of the fluid end 102 including the cutout region 600 having a sinusoidal pattern 1000, which may also be referred to as having a curved or arcuate profile. As noted above, while various embodiments may include the squared entry and/or the sloped entry, a variety of different patterns may be included for forming the cutout region 600. In this example, the sinusoidal pattern 1000 extends along the length 702 of the fluid end 102. As noted above, in various embodiments of the present disclosure, stuffing box assemblies 200 are

13

arranged such that one or more components bear against the back side 202 of the fluid end 102 and without extending into the bore 100.

FIGS. 11 and 12 are cross-sectional views of an embodiment of the fluid end 102 including the stuffing box assembly 200 in which the pressure ring 208 is integrally formed with and/or is a continuous portion of the fluid end block 104. In this example, the pressure ring 208 extends axially out of the cut out region 600 and may replace or otherwise incorporate the mating face 604 (FIG. 6) described above with a continuous material layout such that the pressure ring 208 is formed from a common material with respect to the fluid end block 104.

As noted above, the pressure ring 208 includes the pressure ring bore 210, which has a larger diameter (e.g., diameter 224) than the bore diameter 214. In this manner, the shelf 216 may be formed along the fluid end block 104 to receive and support the packing materials 218, as noted above. Accordingly, the packing materials 218 may be described as not extending into the bore 110 of the fluid end block 104 and being axially blocked or restricted from moving into the bore 110 by the shelf 216. Further illustrated in FIGS. 11 and 12 are the seal groove 228 and seal 230 between the pressure ring 208 and the adjacent ring 220.

As will be appreciated, various features of FIG. 2 are shared in FIGS. 11 and 12, including the configuration of the adjacent ring 220 with respect to the pressure ring 208. For example, in this example, a lip 226 of the adjacent ring 220 may extend axially toward the bore 110 such that a portion of the pressure ring 208 radially overlaps at least a portion of the adjacent ring 220. This may be referred to as the overlapped region 222, as noted above, where the pressure ring 208 includes the diameter 224 to accommodate the lip 226. Further shown is the retainer 234 extending into the adjacent ring 220 to secure the packing materials 218 in place.

Various embodiments of the present disclosure illustrate a secure coupling between the adjacent ring 220 and the pressure ring 208 and/or the fluid end block 104 via one or more fasteners 300, which may extend through apertures or holes through the adjacent ring 220 and/or pressure ring 208 and/or fluid end block 104 that are aligned to permit passage of the fasteners 300. As noted above, fasteners 300 are provided by way of example only and various other components may be utilized to join different features together, such as press fittings, interference fits, adhesives, bayonet connections, tongue and groove fittings, and the like.

Embodiments of the present disclosure further illustrate an arrangement that permits easier access to different components of the stuffing box assembly 200 due to the improved clearance between components. For example, as noted above, the pressure ring 208 may extend axially away from the back side 202 such that at least a portion of the pressure ring thickness 262 is external the groove 204. That is, the pressure ring thickness 262 may include the internal portion 264 that is within the groove 204 and the external portion 266 that is outside of the groove 204. In various embodiments, the internal portion 264 is less than the external portion 266, but it should be appreciated that a variety of configurations may be utilized within the scope of the present disclosure (e.g., internal greater than external, portions equal, etc.). When combined with the groove 204, which may have a depth 606 (FIG. 6), additional clearance area may be provided to various components, such as the fasteners 300, thereby providing improved access for maintenance or other operations, such as packing replacement and the like.

14

FIG. 13 is a sectional perspective view of an embodiment of the stuffing box assembly 200 illustrating the pressure ring 208 formed as an integral feature of the fluid end block 104. In this example, the pressure ring 208 extends away from the back side 202, which may permit increased working area for operators, as shown with respect to the illustrated fasteners 300. Further illustrated is the arrangement of the adjacent ring 220 axially outward with respect to the back side 202. As noted, this configuration may permit the adjacent ring 220 to be formed from materials that are less expensive and/or less likely to be in contact with corrosive or otherwise high pressure fluids, for example, due to the arrangement of the packing materials 218. It should, however, be appreciated that various embodiments may adjust one or more dimensions such that the adjacent ring 220 extends axially closer to the back end 202 such that the packing materials 218 overlap or otherwise are positioned adjacent to the adjacent ring 220.

FIG. 14 is a perspective view of an embodiment of the stuffing box assembly 200 that illustrates the adjacent ring 220 extending axially away from the back side 202. Further illustrated in this configuration is the injection port 250. As shown, the fasteners 300 are arranged circumferentially to extend through the adjacent ring 220, which may further go through aligned holes in the pressure ring 208 (FIGS. 11-13), which may or may not extend all the way to the fluid end block 104 (e.g., through the entire pressure ring thickness 262 (FIGS. 11 and 12)).

This application is a continuation of U.S. Non-Provisional application Ser. No. 17/864,873, filed Jul. 14, 2022, titled "DUAL RING STUFFING BOX," which is a continuation-in-part of U.S. patent application Ser. No. 17/213,854, filed Mar. 26, 2021, titled "DUAL RING STUFFING BOX," now U.S. Pat. No. 11,391,374, issued Jul. 19, 2022, which claims the benefit of U.S. Provisional Patent Application No. 63/137,501, filed Jan. 14, 2021, titled "DUAL RING STUFFING BOX," and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,314, filed Feb. 4, 2021, titled "FLUID END FOR A PUMPING SYSTEM," now U.S. Patent No. D933104, issued Oct. 12, 2021, and is a continuation-in-part and claims the benefit of U.S. patent application Ser. No. 29/769,318, filed on Feb. 4, 2021, titled "FLUID END FOR A PUMPING SYSTEM," now U.S. Patent No. D933105, issued Oct. 12, 2021, the disclosures of which are incorporated herein by reference in their entireties for all purposes.

The foregoing disclosure and description of the disclosed embodiments is illustrative and explanatory of the embodiments of the disclosure. Various changes in the details of the illustrated embodiments can be made within the scope of the appended claims without departing from the true spirit of the disclosure. The embodiments of the present disclosure should only be limited by the following claims and their legal equivalents.

The invention claimed is:

1. A fluid end comprising:

- a fluid end body having a base;
- a fluid end bore positioned in the fluid end body and having a fluid end bore diameter;
- a pressure ring extending from the base and having a pressure ring bore, the pressure ring bore having a pressure ring bore diameter larger than the fluid end bore diameter;
- an adjacent ring connected to the fluid end, the pressure ring positioned between the adjacent ring and the base, the adjacent ring including an aperture;

15

one or more fasteners extending through the aperture of the adjacent ring to connect the adjacent ring to the fluid end, the one or more fasteners extending through the pressure ring and into the base; and
 packing material positioned within at least a portion of the pressure ring bore, so that axial movement toward the fluid end is inhibited by the base.

2. The fluid end of claim 1, further comprising:
 a shelf positioned to support the packing material, the shelf located at a transition between the pressure ring bore diameter and the fluid end bore diameter, and wherein axial movement of the packing material further is blocked by the base.

3. The fluid end of claim 1, wherein the pressure ring is integrally formed with the fluid end, and wherein the shelf includes, at least in part, a spacer ring.

4. The fluid end of claim 1, further comprising:
 a cut out region defining a recessed area positioned to surround the pressure ring, the cut out region extending axially into the fluid end face, toward the base, for a cut out depth, and wherein axial movement of the packing material further is blocked by the base.

5. The fluid end of claim 4, wherein the pressure ring has a pressure ring thickness, wherein the pressure ring thickness includes an internal portion and an external portion, and wherein the internal portion has a thickness approximately equal to the cut out depth.

6. The fluid end of claim 4, wherein the cut out region extends for a length of the fluid end body.

7. The fluid end of claim 1, further comprising:
 an overlapping region located between the pressure ring and the adjacent ring, and wherein the pressure ring includes a recess to receive a lip of the adjacent ring.

8. The fluid end of claim 1, further comprising:
 a seal positioned between the pressure ring and the adjacent ring.

9. A fluid end comprising:
 a fluid end body having a base;
 a fluid end bore positioned in the fluid end body and having a fluid end bore diameter;
 a pressure ring extending from the base and having a pressure ring bore, the pressure ring bore having a pressure ring bore diameter larger than the fluid end bore diameter;
 an adjacent ring connected to the fluid end, the pressure ring positioned between the adjacent ring and the base; one or more fasteners extending through an aperture of the adjacent ring to connect the adjacent ring to the fluid end;
 packing material positioned within at least a portion of the pressure ring bore, so that axial movement toward the fluid end is inhibited by the base; and
 a shelf positioned to support the packing material and located at a transition between the pressure ring bore diameter and the fluid end bore diameter.

10. A fluid end comprising:
 a fluid end body;
 a recessed area positioned in the fluid end body;

16

a fluid end bore positioned in the fluid end body;
 a pressure ring extending from the recessed area, the pressure ring having a pressure ring thickness and a pressure ring bore positioned co-axially with the fluid end bore, the pressure ring bore having a diameter greater than the fluid end bore;
 an adjacent ring positioned against the pressure ring and connected to the fluid end body;
 one or more fasteners extending through the pressure ring thickness;
 packing material positioned within the pressure ring bore, at least a portion of the packing material positioned external to the fluid end bore; and
 an injection port extending through the adjacent ring, the injection port positioned in a low pressure region.

11. The fluid end of claim 10, further comprising a seal positioned between the pressure ring and the adjacent ring, wherein the pressure ring and the adjacent ring are connected together via the one or more fasteners, and wherein the injection port further is positioned axially away from the seal.

12. The fluid end of claim 10, wherein the pressure ring is integrally formed with the fluid end.

13. The fluid end of claim 10, wherein the pressure ring and the adjacent ring are at least one of threaded together or connected together via a bayonet connection.

14. The fluid end of claim 10, further comprising:
 a shelf located at a transition between the pressure ring bore and the fluid end bore, and wherein the shelf receives and supports the packing material.

15. The fluid end of claim 10, wherein the recessed area comprises one or more of a polygonal profile or a curved profile.

16. A method for fluid end assembly, the method comprising:
 providing a fluid end including a recessed area extending into a fluid end face and a pressure ring extending axially out of the recessed area;
 positioning an adjacent ring against the pressure ring, the pressure ring having a selected pressure ring thickness and a pressure ring bore;
 securing the adjacent ring to the fluid end via one or more fasteners extending through the selected pressure ring thickness; and
 installing packing material into the pressure ring bore.

17. The method of claim 16, further comprising:
 connecting a retainer, thereby to compress the packing material against a shelf located at a transition between a fluid end bore and the pressure ring bore.

18. The method of claim 16, wherein the pressure ring is integrally formed with the fluid end.

19. The method of claim 16, further comprising:
 positioning a seal between the pressure ring and the adjacent ring, and connecting the pressure ring and the adjacent ring together via one or more fasteners.

20. The method of claim 19, wherein the fluid end further includes an injection port located axially away from the seal.

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